

- 11 (a) amplitude modulation(*allow AM*)
- (b) (i) frequency = $1 / \text{period}$ C1
 = 100 kHz A1 [2]
- (ii) frequency = 10 kHz A1 [1]
- (c) (i) vertical line at 100 kHz B1
 vertical lines at 90 kHz and 110 kHz B1
 lines at 90 kHz and 110 kHz same length and shorter than at 100 kHz B1 [3]
- (ii) 20 kHz B1 [1]
- [Total: 8]**

- 12 (a) (i) base stations B1 [1]